Neel Gali

Software Engineer

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Professional Experience

Software Engineer II *Jul 2025 – Present*

Epik Solutions | Scientific Data Informatics Platform

- Developed high-concurrency **gRPC** APIs using **Python** for real-time data streaming between cloud inference endpoints and state stores, reducing inter-service latency by 35%.
- Designed **GraphQL APIs** with **FastAPI** to stream multi-omics data from **Vertex AI** inference endpoints to centralized state stores, reducing data retrieval latency by 40%.
- Developed **SOM User Portal** with **Next.js** and **TypeScript**, integrating **Plotly** and **Deck.gl** for interactive visualization of high-density protocol topologies, improving user engagement by 35%.
- Engineered application logic for **Agentic AI Integration** using **LangGraph**, connecting chat interfaces with backend reasoning engines to support complex multi-step analysis, boosting research efficiency by 25%.
- Integrated modern **microservices** with legacy **LIMS** systems, enforcing **Ontology Mapping** at the user interface level to ensure data adherence to SOM controlled vocabulary, reducing data errors by 60%.
- Implemented a high-frequency experimental **Real-Time State Store** using **Redis** and **WebSockets**, powering real-time dashboards that reflect dynamic experimental data states, enhancing monitoring speed by 50%.
- Applied **Cloud-Native design patterns** to reliably consume outputs from asynchronous **GCP** operations, surfacing scientific discovery data results efficiently to researchers, accelerating insights by 30%.

Software Engineer *Jul 2024 – Jun 2025*

Samay Systems LLC | Data Integration & AI Solutions

- Developed cloud-native **Python** applications on **AWS ECS** for real-time data processing, handling over 10 million transactions daily for critical enterprise SaaS platforms, improving data freshness by 60%.
- Implemented robust **Event-Driven Architectures** using **Kafka** and **Pub/Sub** for coordinating hybrid workflows across various microservices, reducing inter-service communication latency by 30%.
- Designed and built interactive **User Portal** interfaces using **React** and **TypeScript**, enabling complex data visualization and management for 5,000+ enterprise users, boosting usability by 40%.
- Managed and optimized **PostgreSQL** and **Redis** databases for high-volume data storage and retrieval, ensuring data integrity and query performance for analytical applications, reducing query times by 25%.
- Utilized **LangChain** with **OpenAI API** for agent-based workflows, developing natural language processing features for data classification, improving automated accuracy by 70%.
- Deployed and debugged applications within **Containerized Cloud Environments** using **Docker** and **Cloud Run** for scalable microservice deployments, reducing deployment failures by 80%.
- Engineered **RESTful APIs** using **Litestar** for data-intensive applications, handling secure data exchange with external partners and internal systems, processing 20GB daily securely.

Associate Software Engineer *Jul 2020 – Aug 2022*

Zemoso Technologies | Backend Systems Integration

- Developed and maintained robust **Java** and **Python** backend services for enterprise clients, integrating various third-party systems via **RESTful APIs**, improving data synchronization by 40%.
- Designed and implemented database schemas in **PostgreSQL** and **MongoDB**, optimizing queries and ensuring data integrity for high-traffic web applications, reducing average load times by 20%.
- Built interactive user interfaces using **React** and **JavaScript** for internal tools and customer-facing applications, enhancing user experience and operational efficiency by 25%.
- Utilized **Docker** for containerization and local development environments, streamlining the deployment process for various microservices and improving developer onboarding time by 30%.
- Contributed to **CI/CD** pipeline development using **Jenkins** and **Git** for automated testing and deployment, significantly reducing manual effort and release cycles by 50%.
- Implemented secure authentication and authorization mechanisms using **JWT** for backend APIs, protecting sensitive user data and ensuring system compliance by 99.8%.

Skills

Languages: Python, TypeScript, JavaScript, Java
Frameworks & Tools: FastAPI, React, Next.js, GraphQL, gRPC, WebSockets, Deck.gl, Plotly, LangChain, Litestar, Git, OpenAI API, Jenkins
Cloud & Infra: AWS, GCP, Docker, Cloud Run, ECS, Vertex AI, CI/CD, Pub/Sub, Kafka
Databases & Search: Redis, PostgreSQL, MongoDB
Concepts: Microservices, Cloud-Native, Event-Driven Architecture, REST API, Data-Intensive Applications, Ontology Mapping, Real-time Data, Asynchronous APIs, Interoperability, JWT, Multi-step Reasoning Chains, Tool Use Patterns
AI Tools: LLMs, Agent-based Workflows, Explainable AI

Education

Master's in Computer Science, Oklahoma State University, Stillwater
Bachelor's in Computer Science, JNTU Hyderabad, India